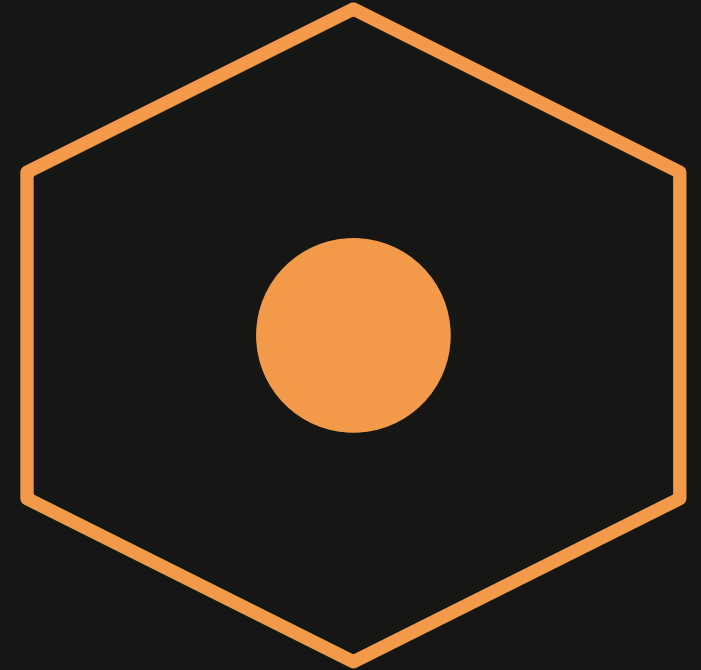


FieldMind

Memory for field technicians that works with no network, and knows what it is allowed to share.



Team Compilers

github.com/yabhinav1/fieldmind

The most useful field knowledge is created where the network is worst

1

No signal

Plant floors, basements and remote sites drop the connection for hours.

2

Mixed notes

One note can hold a fault the fleet needs and a name that must not leave.

3

Drift

Devices that work apart come back with different versions of the truth.

The device that keeps working

1

Record

Write what you saw. The device shows what it will share before you save.

2

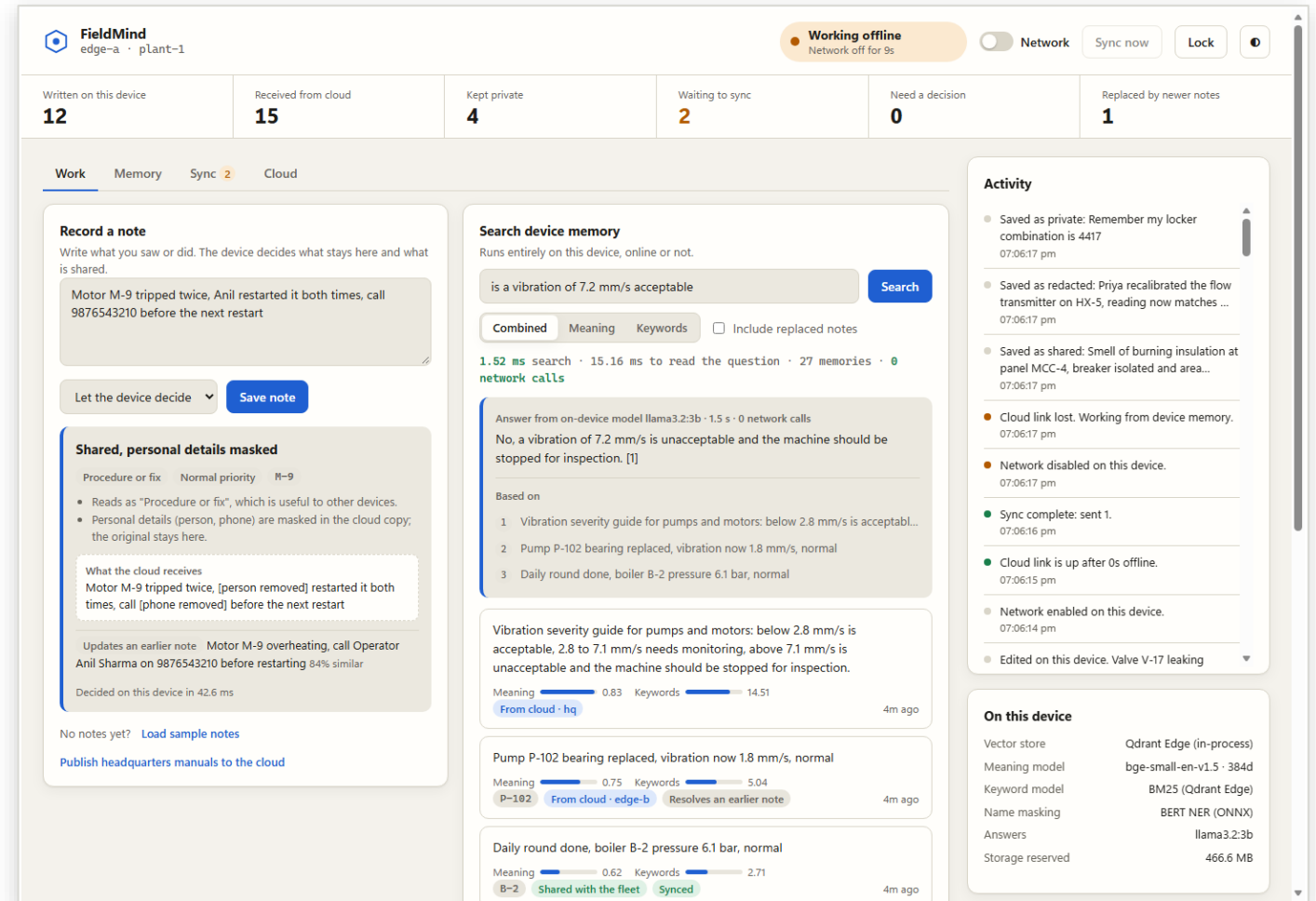
Find

Search your notes and headquarters manuals in about a millisecond.

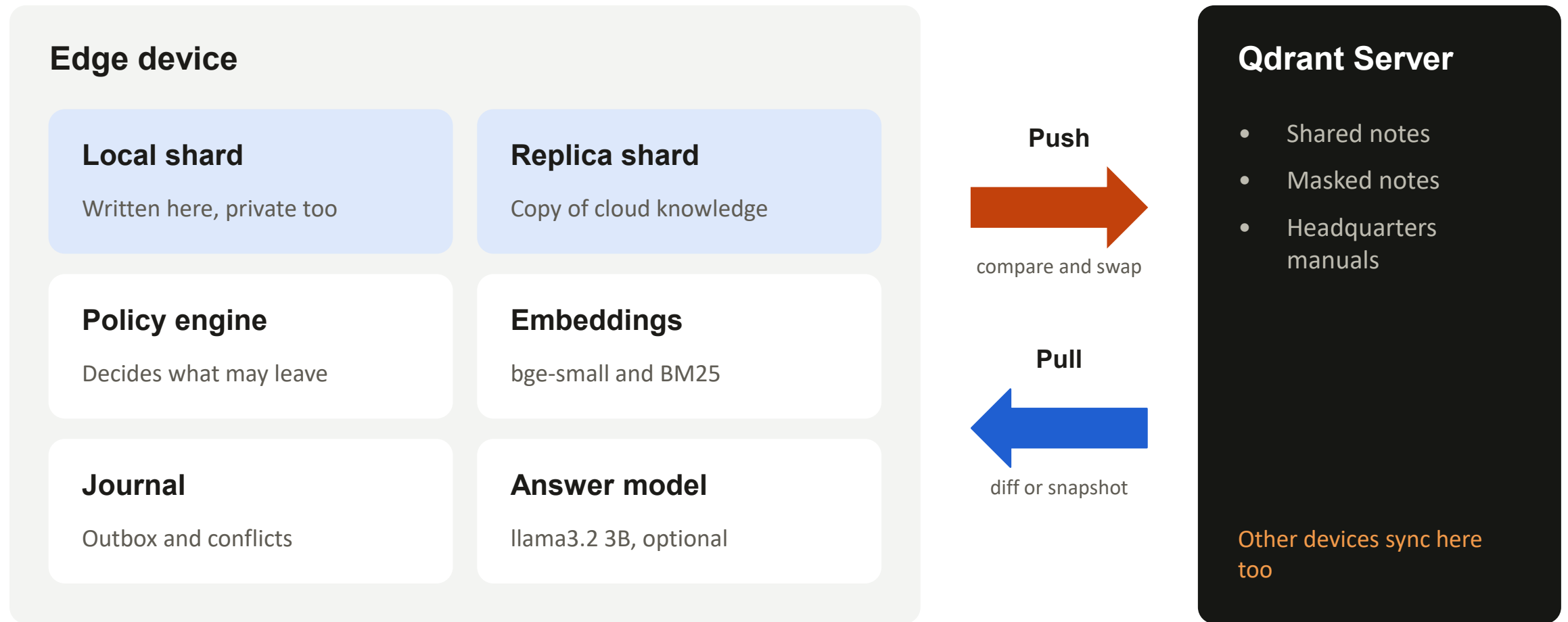
3

Ask

Get a cited answer written on the device. No cloud call.



One device process, one cloud collection



The device decides what leaves

What the technician wrote	What happens
Pump P-102 bearing vibration high at 7.2 mm/s	Shared
Smell of gas near compressor K-4	Shared, urgent, sent first
Suresh and Priya replaced the coupling, call 9876543210	Shared, names and number masked
Technician Ravi Kumar reported chest pain	Private, stays on the device
SCADA panel login password is ...	Private, cannot be overridden

Decided in about 20 ms on the device by a name recognition model, pattern detectors and a semantic classifier. Vectors for a masked note are computed from the masked text.

When two devices disagree

1

Nothing is overwritten

An upload only lands if the cloud is still at the version the device last saw.

2

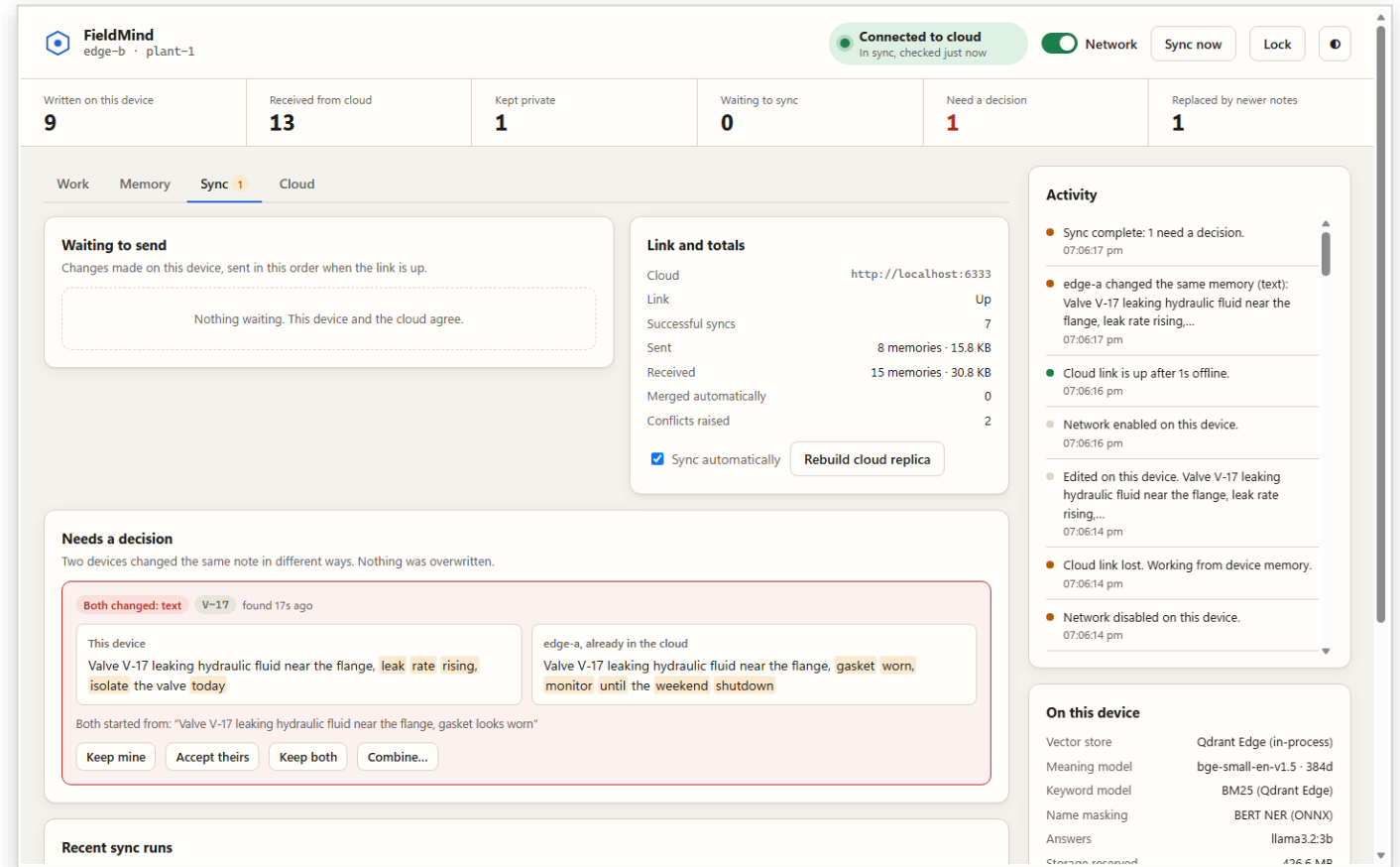
Most edits merge

Changes to different fields combine automatically.

3

A person settles the rest

Keep mine, accept theirs, keep both or combine. Even while offline.



What Qdrant does for us

Qdrant feature	How FieldMind uses it
Edge shards, in process	Two per device: local notes and a cloud replica
Built-in BM25 with IDF	Keyword half of hybrid search, with no server
Dense vectors and filters	Meaning search narrowed by asset, scope and status
Shard snapshots	A device far behind restores its replica from one file
Conditional upsert	Compare and swap, so devices never clobber each other
One schema on both sides	The same dense and sparse vectors on edge and server

qdrant-edge-py 0.8.0 with Qdrant Server 1.19.1

Measured, not estimated

1 ms

hybrid search across both shards

0

network calls to search or answer

20 ms

to decide what may leave the device

1–3 s

for a cited answer from a 3B model

57

automated tests, two devices each

30

checks in the live end-to-end run

Ryzen 7 7435HS laptop, RTX 3050 4 GB. Also verified on Linux through a real network disconnect.

Five moments in the demo

- 1 Switch the network off and keep working
- 2 Ask a question, get a cited answer
- 3 Watch names get masked before upload
- 4 Reconnect: urgent notes go first
- 5 Two devices disagree, a person decides

Next

- Encrypt notes at rest
- TLS for remote dashboards
- Name masking beyond English